Project Initialization and Planning Phase

Date	10 July 2026
Team ID	
Project Title	Smart Lender – Loan Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed Smart Lender – Loan Prediction System uses Machine Learning to predict loan eligibility based on applicant details. The system helps financial institutions make faster, more accurate, and reliable loan approval decisions while reducing manual effort and improving customer experience.

Project Overview	
Objective	The primary objective of this project is to develop a Machine Learning-based loan prediction system that accurately predicts whether a loan applicant is eligible for approval based on financial and personal information.
Scope	The system analyzes applicant information such as income, education, employment status, loan amount, credit history, and other financial attributes to predict loan eligibility through a web-based application.
Problem Statement	
Description	Traditional loan approval processes are time-consuming, require manual verification, and may lead to inconsistent decisions. Applicants often experience delays and uncertainty during the approval process.
Impact	Implementing an intelligent loan prediction system improves decision accuracy, reduces processing time, minimizes manual effort, and enhances customer satisfaction.

Proposed Solution	
Approach	Develop a Machine Learning model using historical loan data and deploy it as a Flask web application where users can enter applicant details and instantly receive loan eligibility predictions.
Key Features	• Machine Learning-based loan eligibility prediction. • User-friendly web interface using HTML and CSS. • Fast and accurate prediction results. • Uses multiple ML algorithms (Decision Tree, Random Forest, KNN, XGBoost). • Real-time prediction through Flask. • Easy deployment and maintenance.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU	intel Core i5 or above
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv